

Healthmate — Architecture Overview

Project Overview

Healthmate is a multi-agent health assistant built with **LangChain** and **Google Gemini**. It routes user health queries to specialized AI agents, each equipped with domain-specific tools — from body metric calculators to medical PDF analyzers.

System Architecture

High-Level Flow

User Input → **Leader Router** → Agent(s) → Tools → Response

The Leader Router acts as a keyword-based dispatcher. It reads the user's message, matches it against predefined keyword lists, and invokes the appropriate agent(s). Multiple agents can be triggered by a single input if it contains keywords from both categories.

Components

1. Leader Router

`Agents/leader.py`

The entry point for all user input. Uses keyword matching to decide which agent(s) to call.

- **Calc keywords:** `bmi`, `body fat`, `waist`, `hip`, `bri`, `ratio`
- **Medical keywords:** `pain`, `headache`, `drug`, `medicine`, `fever`, `symptom`, `.pdf`, `blood test`

Both agents can be triggered simultaneously if the input contains keywords from both lists. Results are concatenated and returned together.

2. Agents

Calc Agent

`Agents/calc.py`

Powered by **Gemini 2.5 Flash Lite** (fast, low-cost). Handles all body measurement and composition calculations.

Tool	Description
<code>bmiCalculator</code>	Calculates BMI and weight category
<code>waistToHeightRatio</code>	Computes WHtR with health category
<code>waistToHipRatio</code>	Computes Waist-to-Hip Ratio
<code>bodyAdiposityIndex</code>	Estimates body fat via hip/height formula
<code>bodyFatPercentage</code>	US Navy method (male & female formulas)
<code>bodyRoundnessIndex</code>	Elliptical body shape index

Medical Agent

`Agents/medical.py`

Powered by **Gemini 2.5 Flash** (more capable). Handles drug lookups, symptom checking, and lab report analysis.

Tool	Description
<code>drugInfo</code>	Fetches real drug data from DailyMed (NIH)
<code>symptomChecker</code>	Returns possible conditions based on reported symptoms
<code>medicalTestAnalyzer</code>	Parses medical PDFs, flags out-of-range values, and explains them in plain English

3. Tools

`Tools/bodyTools.py`

Six pure-Python calculators. Each accepts a key=value string as input and returns a result with a disclaimer. No external API calls.

`Tools/drugs.py`

Calls the **DailyMed REST API** (NIH) to fetch official drug labeling. Returns the drug name, source, and up to 2,000 characters of label text including uses, warnings, and allergen information.

Tools/symptom.py

A symptom checker covering common conditions including headache, back pain, fever, cough, and sore throat. Returns a list of possible conditions based on the reported symptoms.

Tools/pdf.py

Extracts text from medical PDF reports using **PyMuPDF (fitz)**, compares values against a `normalRanges` dictionary, and uses Gemini to explain any abnormal findings in plain, easy-to-understand English.

Covers:

- Blood panel (Hemoglobin, WBC, Platelets, Glucose, Creatinine)
- Liver function (ALT, AST, ALP, Bilirubin, Albumin, Total Protein)
- Kidney function (BUN, Uric Acid, Sodium, Potassium, Chloride, Calcium, eGFR)
- Urine (pH, Specific Gravity)

Also supports aliases (e.g. `HGB` → `Hemoglobin`, `SGPT` → `ALT`).

4. Memory

Memory/memory.py

Uses LangChain's `ConversationBufferMemory` to maintain chat history across turns. Also defines a **User Profile** that is injected into the system prompt:

```
name, age, weight_kg, height_m, allergies
```

The Medical Agent uses this profile to warn about allergens when discussing drugs.

5. Configuration

config.py

Loads the Gemini API key from `key.env` and initializes three LLM instances:

Instance	Model	Temperature	Used By
<code>leaderLLM</code>	gemini-2.5-flash-lite	0	Leader Router
<code>calcLLM</code>	gemini-2.5-flash-lite	0	Calc Agent

Instance	Model	Temperature	Used By
<code>medicalLLM</code>	gemini-2.5-flash	0.3	Medical Agent

Dummy Test Data

`Dummy tests/`

Includes 8 sample PDF reports (normal and mixed results) for:

- Blood tests
- Liver function tests
- Kidney function tests
- Urine tests

Also includes Python scripts to generate these PDFs programmatically.

Tech Stack

Layer	Technology
LLM Provider	Google Gemini (via <code>langchain-google-genai</code>)
Agent Framework	LangChain (zero-shot-react-description)
PDF Parsing	PyMuPDF (<code>fitz</code>)
Drug Data	DailyMed REST API (NIH)
Memory	LangChain <code>ConversationBufferMemory</code>
Env Management	<code>python-dotenv</code>